EDS223 - HW2

Exploring patterns of environmental justice

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Setup

```
# Load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(here)
```

```
here() starts at /Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                   from
  st_interpolate_aw.stars sf
```

```
library(spData)
library(knitr)
library(ggpubr)

# Set warnings to display as they occur
options(warn = 1)

# Set ggplot theme
set_theme(theme_bw())
```

```
# Load data
# EJScreen data - environmental and demographic US info at the block group level
ejscreen <- sf::st_read(here("data","ejscreen",
                             "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# HOLC Redlining data - HOLC maps and area descriptions for LA
holc <- sf::st_read(here("data","mapping-inequality",
                             "mapping-inequality-los-angeles.json")) |>
        # Correct invalid geometries to avoid error in st_join when using CRS 4326
        st_make_valid()
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2/data/mapping-inequality/mapping-inequality-los-angeles.json'
```

```
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
# Bike + Ped Crash data - LA crash records
crashes <- read_csv(here("data","SWITRS_LA","Crashes.csv")) |>
    # Remove rows w/ missing coordinates so convert to sf is possible
    # Use POINT_X,POINT_Y each with only 3% missing data (Lat/Long each have 38%)
    drop_na(POINT_X,POINT_Y) |>
    # Convert crashes csv to sf data - SWITRS uses CRS WGS 1984 EPSG:4326
    st_as_sf(coords = c("POINT_X","POINT_Y"), crs = 4326)
```

```
Rows: 251100 Columns: 80
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (45): COLLISION_TIME, OFFICER_ID, REPORTING_DISTRICT, CHP_BEAT_TYPE, CI...
dbl  (33): CASE_ID, ACCIDENT_YEAR, JURIS, DAY_OF_WEEK, CHP_SHIFT, POPULATION...
date  (2): PROC_DATE, COLLISION_DATE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Warning - CRS - confirm crs match, transform if not
if ((st_crs(ejscreen)$epsg == st_crs(holc)$epsg) &
     (st_crs(ejscreen)$epsg == st_crs(crashes)$epsg) &
     (st_crs(holc)$epsg == st_crs(crashes)$epsg)){
    # All three CRS match - true
    print("CRS do match!")
} else {
    # CRS match - false
    warning("CRS do NOT match!")
    print("Changing EJScreen data CRS to match HOLC and Crashes data CRS EPSG:4326...")
    # CRS transforms to match ejscreen data
    ejscreen <- st_transform(ejscreen, crs = st_crs(holc)$epsg)
    crashes <- st_transform(crashes, crs = st_crs(holc)$epsg)
    print("CRS match complete!")
}
```

```
Warning: CRS do NOT match!
```

```
[1] "Changing EJScreen data CRS to match HOLC and Crashes data CRS EPSG:4326..."
[1] "CRS match complete!"
```

Part 1: Legacy of redlining in current environmental (in)justice

Question 1: Map of Historical Redlining Neighborhoods

```
# Correct HOLC colors per grade
holc_colors = c("A" = "#76a865",
                "B" = "#7cb5bd",
                "C" = "#f1db11",
                "D" = "#d9838d",
                "NA" = "#000000")
# Create holc polygon map over la county basemap
tm_basemap(server = "Esri.WorldTopoMap") +
tm_shape(holc) +
    tm_polygons(fill = "grade",
                fill_alpha = 0.5,
                lwd = 0.5,
                fill.scale = tm_scale_categorical(values = holc_colors),
                fill.legend = tm_legend(title = "HOLC Grade",
                                        # Specify legend layout
                                        position = tm_pos_out("right"))) +
tm_compass(position = c("left","bottom")) +
tm_scalebar(position = c("left","bottom")) +
tm_options(component.autoscale = FALSE) +
tm_title("Los Angeles County: Redlined Neighborhoods",
            frame = TRUE,
            size = 1.5) +
tm_layout(# Specify legend title size
          legend.title.size = 1.0,
          # Specify legend text size
          legend.text.size = 0.8,
          # Set font family
          text.fontfamily = "sans",
          # Set inner margins to give scalebar room
          inner.margins = c(0.1,0.1,0.05,0.05)
          )
```

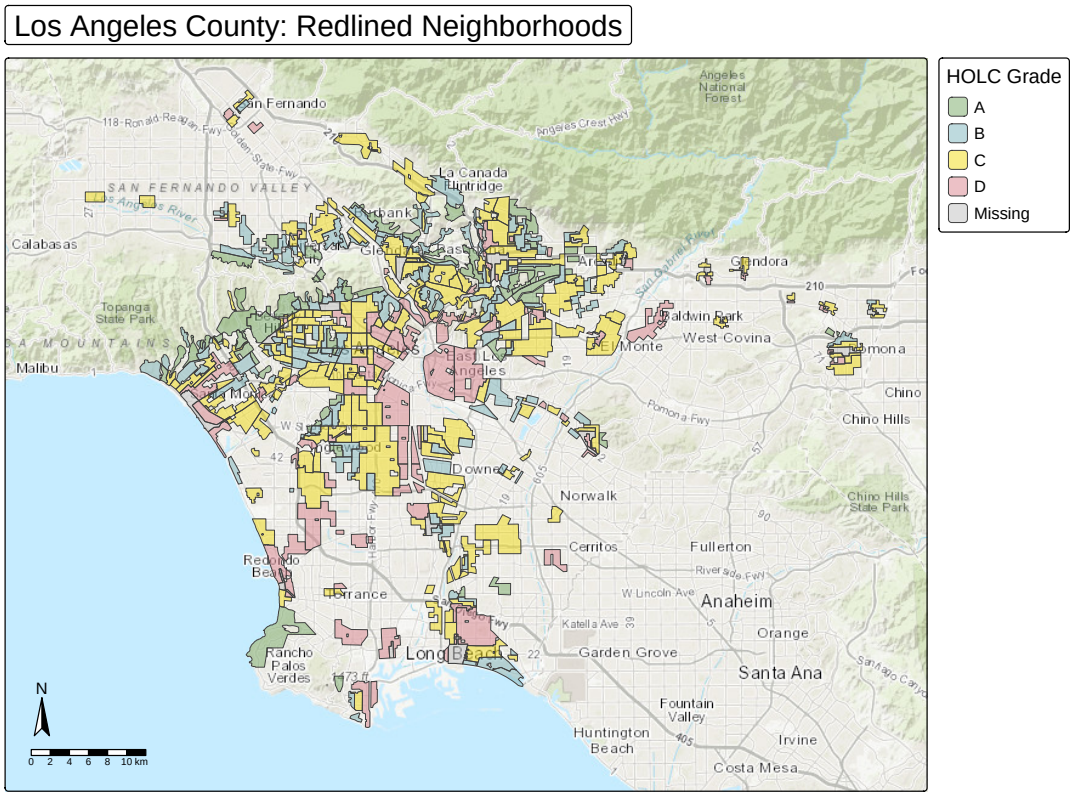


Figure 1: Map of Los Angeles (LA) county in California (CA). Neighborhoods with HOLC grades assigned from 1935-1940 - A, B, C, and D - are represented on this map by color. An "A" HOLC grade is the best "residential security", while a "D" HOLC grade is the worst "residential security."

Question 2: Table of Census Blocks Within HOLC Grades

```
# Filter ejscreen to los angeles county in CA
los_angeles_ejs <- ejscreen |>
    dplyr::filter((ST_ABBREV == "CA") &
                (CNTY_NAME == "Los Angeles County"))

# Confirm st_intersect will work in st_join for ejscreen LA and holc datasets
if(any(st_intersects(los_angeles_ejs, holc, sparse = FALSE))){
  print("st_intersect will work - geometries overlap.")
} else{
```

```
  stop("st_intersect will not work - no overlapping geometries.")
}
```

```
[1] "st_intersect will work - geometries overlap."
```

```
# Join holc to los_angeles_ejs by intersecting geometry
la_holc_join <- st_join(los_angeles_ejs, holc, join = st_intersects) |>
    # Select columns relevant to Part 1
    select(ID,grade,LOWINCPCT,P_PM25,P_LIFEEXPPCT) |>
    # Drop geometries
    st_drop_geometry()

# Warning - Join - check if number of rows matches original data (left join)
if(nrow(los_angeles_ejs) == nrow(la_holc_join)){
  print("Join matches original data dimensions of los_angeles_ejs")
} else{
  warning("Join does not match original data dimensions of los_angeles_ejs.")
  print(paste("los_angeles_ejs has", nrow(los_angeles_ejs), "rows"))
  print(paste("la_holc_join has", nrow(la_holc_join), "rows"))
  print("Duplicated rows are census blocks intersected by 2+ grades.")
  print("Highest grade for such census blocks will be kept.")
  print("See Results Summary for details.")
}
```

```
Warning: Join does not match original data dimensions of los_angeles_ejs.
```

```
[1] "los_angeles_ejs has 6591 rows"
[1] "la_holc_join has 8988 rows"
[1] "Duplicated rows are census blocks intersected by 2+ grades."
[1] "Highest grade for such census blocks will be kept."
[1] "See Results Summary for details."
```

```
# Check for duplicates - 1731 census blocks overlapping 2+ HOLC grades
la_holc_dups <- la_holc_join |>
    group_by(ID) |>
    summarise(Num_Grades = n()) |>
    filter(Num_Grades >= 2)

# Keep only "highest" grade row for each census block ID w/ 2+ grades
la_holc <- la_holc_join |>
```

```
    # Sort by ID, then HOLC grade
    arrange(ID, grade) |>
    # Keep only first row per ID
    distinct(ID, .keep_all = TRUE)

# Warning - Join - confirm duplicate fix worked
if(nrow(los_angeles_ejs) == nrow(la_holc)){
  print("After removing duplicates, join matches original data dimensions!")
} else{
  stop("Join still does not match original data dimensions of los_angeles_ejs.")
}
```

```
[1] "After removing duplicates, join matches original data dimensions!"
```

```
# Create summary table of census block (cb) counts and percentages per holc grade
la_holc_graden <- la_holc |>
    group_by(grade) |>
    summarise(cb_count = n()) |>
    mutate(cb_perc = (cb_count / nrow(la_holc)) * 100)

# Use kable to format table output
kable(la_holc_graden,
        col.names = c("HOLC Grade","Census Blocks per Grade (Count)",
                      "Census Blocks per Grade (Percent)"))
```

Table 1: Counts and percentages of EJScreen census blocks per HOLC Grade in LA county. The percentages were calculated using a total LA county census block total of 6591 (see Results Summary for details).

HOLC Grade	Census Blocks per Grade (Count)	Census Blocks per Grade (Percent)
A	386	5.856471
B	879	13.336368
C	1914	29.039600
D	773	11.728114
NA	2639	40.039448

Question 3: EJScreen Data Visualizations Within HOLC Grades

```r
# Create df for visualizations
la_holc_sum <- la_holc |>
    # Group by grade
    group_by(grade) |>
    # Calculate means by grade
    summarise(mean_LOWINCPCT = mean(LOWINCPCT, na.rm = TRUE),
              mean_P_PM25 = mean(P_PM25, na.rm = TRUE),
              mean_P_LIFEEXPPCT = mean(P_LIFEEXPPCT, na.rm = TRUE)) |>
    rename("HOLC Grade" = "grade")
```

```r
# Generate plots

# Generate bar plot - mean_LOWINCPCT across HOLC grades
sum_plot1 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_LOWINCPCT*100,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Low Income",
         y = "Mean %")

# Generate bar plot - mean_P_PM25 across HOLC grades
sum_plot2 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_P_PM25,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Particulate Matter 2.5",
         y = "Mean US Percentile")

# Generate bar plot - mean_P_LIFEEXPPCT across HOLC grades
sum_plot3 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_P_LIFEEXPPCT,
        fill = `HOLC Grade`)) +
```

```
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Life Expectancy",
         y = "Mean US Percentile")

# Arrange bar plots in one figure
ggarrange(sum_plot1, sum_plot2, sum_plot3, ncol = 3, nrow = 1,
            labels = c("(a)", "(b)", "(c)"),
            font.label = list(face = "plain"),
            common.legend = TRUE,
            legend = "bottom") |>
# Add figure title
annotate_figure(top = text_grob("HOLC Neighborhoods: Current EJScreen Statistics",
                                size = 20))
```

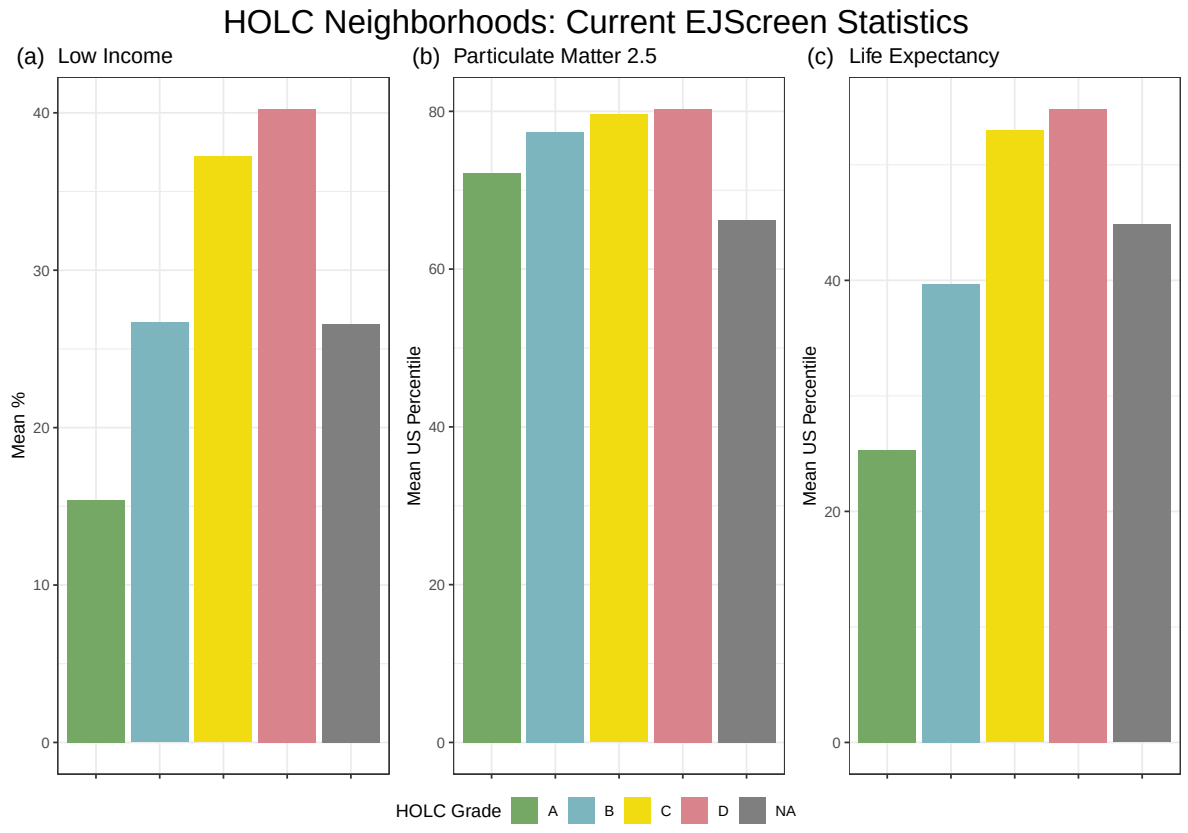


Figure 2: Bar plots showing current (2023) EJScreen data for three statistics - (a) low income

(%), (b) US percentile for Particulate Matter (PM) 2.5, and (c) US percentile for low life expectancy - each averaged across historical HOLC neighborhood (redlining) grades.

Question 4: Results Summary

For LA county, the number of EJScreen census blocks within a Home Owners' Loan Corporation (HOLC) grade is not evenly distributed across the four grades - "A", "B", "C", "D". The "A" and "B" grades combined only have around 19% of the total LA county census blocks noted in EJScreen, while grade "C" has about 29% and grade "D" has 11% (32% of the census blocks were not associated with a grade) (*Table 2*). It should be noted that the total number of EJScreen census blocks in LA county used to calculate percentages was 6591. After combining the EJScreen and HOLC data, 1731 census blocks intersected with multiple HOLC grades - in these cases, only the highest grade was kept per census block.

When comparing the HOLC grade means of three metrics recorded per census block in EJScreen - low income (%), US percentile for Particulate Matter (PM) 2.5, and US percentile for low life expectancy - a similar trend was observed across all three comparisons. The means calculated for low income (%), percentile for PM2.5, and percentile for low life expectancy all increase sequentially from grade "A" to grade "D" (*Figure 2*). This implies that historical redlining by HOLC from 1935-1940 is possibly correlated with current (2023 EJScreen) trends in low income, PM2.5, and low life expectancy. In other words, the census blocks located in grade "A" HOLC neighborhoods currently have the lowest mean percentage of low income households (15.39%), and fall in lower mean percentiles than the other three grades for annual PM2.5 air exposure (~72th percentile) and low life expectancy (~25th percentile). In contrast, grade "D" HOLC neighborhoods have the highest mean percentage of low income households (40.26%), and fall in higher mean percentiles than the other three grades for annual PM2.5 air exposure (~80th percentile) and low life expectancy (~55th percentile). An alternative interpretation of the US percentile metric is that if a group - like grade "D" - is in the 55th US percentile for low life expectancy, then only 45% of the US has a higher low life expectancy metric. Given these results, it appears that historically redlined "C" and "D" grade census blocks do currently experience more negative environmental and socioeconomic conditions than census blocks in historical HOLC "A" and "B" grade neighborhoods.

Part 2: Legacy of redlining in pedestrian and bike accidents

```
# Filter crashes to only Y PED accidents
ped_crashes_filt <- crashes |>
    # Select only relevant columns to Part 2 (geometry sticks)
    select(CASE_ID, PEDESTRIAN_ACCIDENT, COLLISION_SEVERITY) |>
    # Apply filter
    filter(PEDESTRIAN_ACCIDENT == "Y")
```

```r
# Filter crashes to only Y BIKE accidents
bike_crashes_filt <- crashes |>
    # Select only relevant columns to Part 2 (geometry sticks)
    select(CASE_ID, BICYCLE_ACCIDENT, COLLISION_SEVERITY) |>
    # Apply filter
    filter(BICYCLE_ACCIDENT == "Y")

# Confirm st_intersect will work in st_join for holc and crashes_filt datasets
if((any(st_intersects(holc, ped_crashes_filt, sparse = FALSE))) &
    (any(st_intersects(holc, bike_crashes_filt, sparse = FALSE)))){
  print("st_intersect will work - geometries overlap.")
} else{
  stop("st_intersect will not work - no overlapping geometries.")
}
```

```
[1] "st_intersect will work - geometries overlap."
```

```r
# Join holc to ped_crashes by intersecting geometry
holc_ped_crash <- st_join(ped_crashes_filt, holc, join = st_intersects) |>
    # Select columns relevant to Part 2
    select(grade, CASE_ID, PEDESTRIAN_ACCIDENT, COLLISION_SEVERITY) |>
    # Drop geometries
    st_drop_geometry()

# Join holc to bike_crashes by intersecting geometry
holc_bike_crash <- st_join(bike_crashes_filt, holc, join = st_intersects) |>
    # Select columns relevant to Part 2
    select(grade, CASE_ID, BICYCLE_ACCIDENT, COLLISION_SEVERITY) |>
    # Drop geometries
    st_drop_geometry()

# Check for duplicates - none, all IDs have 1 grade match
ped_crash_flag <- holc_ped_crash |>
    group_by(CASE_ID) |>
    summarise(Num_Grade_matches = n()) |>
    arrange(Num_Grade_matches, ascending = FALSE)
bike_crash_flag <- holc_bike_crash |>
    group_by(CASE_ID) |>
    summarise(Num_Grade_matches = n()) |>
    arrange(Num_Grade_matches, ascending = FALSE)
```

```
# Warning - Join - check if number of rows matches original data (left join)
if((nrow(holc_ped_crash) == nrow(ped_crashes_filt)) &
    (nrow(holc_bike_crash) == nrow(bike_crashes_filt))){
  print("Joins match data dimensions of ped_crashes_filt and bike_crashes_filt.")
} else{
  warning("Joins do not match original data dimensions.")
}
```

```
[1] "Joins match data dimensions of ped_crashes_filt and bike_crashes_filt."
```

```
# Find total number of fatalities - ped
num_total_ped_fatalities <- holc_ped_crash |>
    filter(COLLISION_SEVERITY == 1.00) |>
    nrow()
# Find total number of fatalities - bike
num_total_bike_fatalities <- holc_bike_crash |>
    filter(COLLISION_SEVERITY == 1.00) |>
    nrow()

# Create ped summary df
sum_ped_crash <- holc_ped_crash |>
    # Group by HOLC grade
    group_by(grade) |>
    summarise(
        # Count ped accidents per grade
        num_Ped_Acc = n(),
        # Percent ped accidents per grade
        perc_Ped_Acc = num_Ped_Acc/nrow(holc_ped_crash) * 100,
        # Count total fatalities per grade
        # Each row where COLLISION_SEVERITY == 1.00 is TRUE equals 1
        # Sum those 1s to get a count of rows with a value of TRUE
        num_Ped_Fatal = sum(COLLISION_SEVERITY == 1.00),
        # Percent total fatalities per grade
        perc_Ped_Fatal = num_Ped_Fatal/num_total_ped_fatalities * 100
    ) |>
    rename(`HOLC Grade` = grade)

# Create bike summary df
sum_bike_crash <- holc_bike_crash |>
    # Group by HOLC grade
    group_by(grade) |>
    summarise(
```

```
        # Count total ped accidents per grade
        num_Bike_Acc = n(),
        # Percent ped accidents per grade
        perc_Bike_Acc = num_Bike_Acc/nrow(holc_bike_crash) * 100,
        # Count total fatalities per grade
        # Each COLLISION_SEVERITY == 1.00 is TRUE = 1
        # Sum those 1s to get a count of rows with a value of TRUE
        num_Bike_Fatal = sum(COLLISION_SEVERITY == 1.00),
        # Percent total fatalities per grade
        perc_Bike_Fatal = num_Bike_Fatal/num_total_bike_fatalities * 100
    ) |>
    rename(`HOLC Grade` = grade)
```

```
# Plot visualization

# Plot 1: Pedestrian Accidents
plt1 <- ggplot(sum_ped_crash,
            aes(x = `HOLC Grade`,
                y = perc_Ped_Acc,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank()) +
            ylim(0,60) +
            labs(title = "Pedestrian Accidents",
                 y = "Accidents (%)")

# Plot 2: Bike Accidents
plt2 <- ggplot(sum_bike_crash,
            aes(x = `HOLC Grade`,
                y = perc_Bike_Acc,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank(),
                  axis.title.y  = element_blank()) +
            ylim(0,60) +
            labs(title = "   Bicycle Accidents")

# Plot 3: Pedestrian Accidents
```

```r
plt3 <- ggplot(sum_ped_crash,
            aes(x = `HOLC Grade`,
                y = perc_Ped_Fatal,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank()) +
            labs(title = "Pedestrian Fatalities",
                 y = "Fatalities (%)")

# Plot 4: Bike Accidents
plt4 <- ggplot(sum_bike_crash,
            aes(x = `HOLC Grade`,
                y = perc_Bike_Fatal,
                fill = `HOLC Grade`)) +
            geom_bar(stat = "identity") +
            scale_fill_manual(values = holc_colors) +
            theme(axis.title.x = element_blank(),
                  axis.text.x  = element_blank(),
                  axis.title.y  = element_blank()) +
            labs(title = "    Bicycle Fatalities")

# Arrange bar plots in one figure
ggarrange(plt1, plt2, plt3, plt4, ncol = 2, nrow = 2,
            labels = c("(a)", "(b)", "(c)", "(d)"),
            font.label = list(face = "plain"),
            common.legend = TRUE,
            legend = "bottom") |>
# Add figure title
annotate_figure(
    top = text_grob("2020-2025 Pedestrian and Bicycle Accidents Across HOLC Grades",
                    size = 20))
```

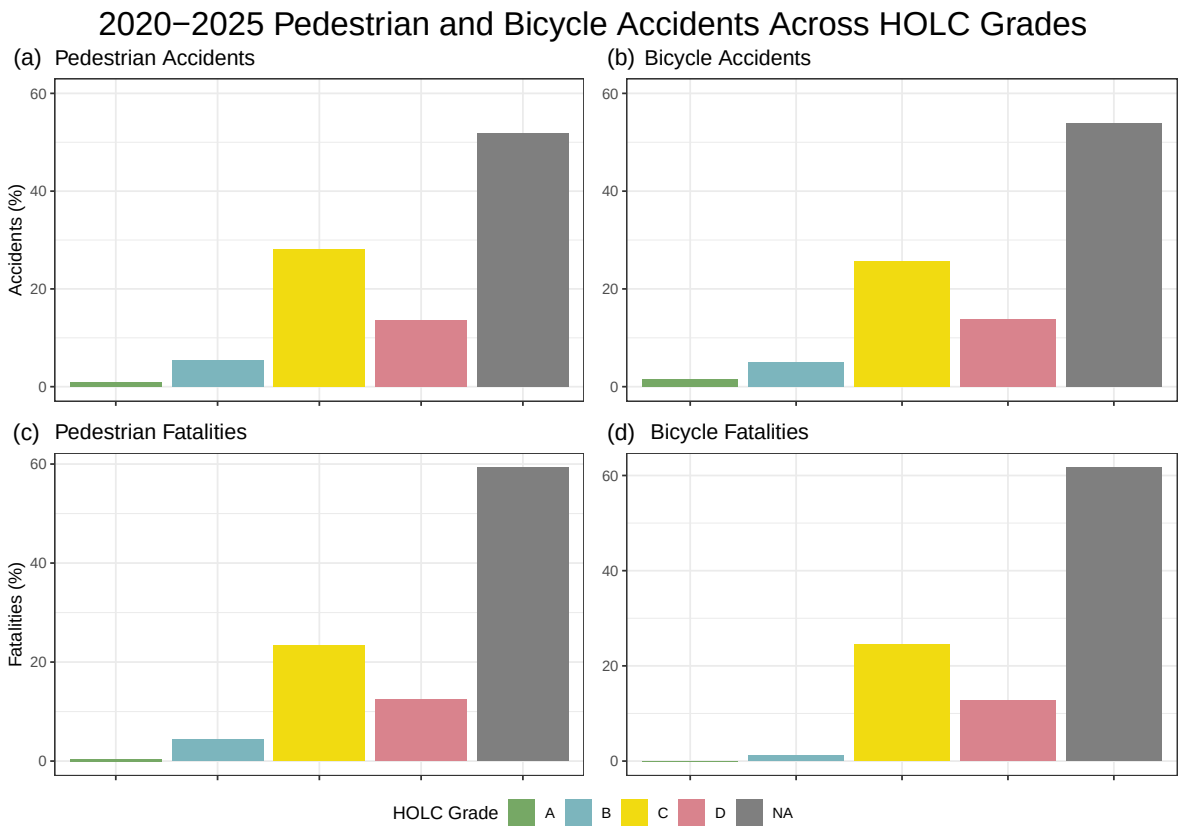


Figure 3: Bar graphs comparing percent pedestrian and bicycle accidents and fatalities from 2020-2025 in LA County across historical HOLC grades. (a) Percent pedestrian accidents (number of pedestrian accidents in grade/total number of pedestrian accidents), (b) percent bicycle accidents (number of bicycle accidents in grade/total number of bicycle accidents), (c) percent pedestrian fatalities (number of pedestrian fatalities in grade/total number of pedestrian fatalities), (d) percent bicycle fatalities (number of bicycle fatalities in grade/total number of bicycle fatalities).

Results Summary

While many LA county accidents from 2020-2025 occurred outside of historic HOLC grade neighborhoods - represented by the relatively higher percentages of "NA" grade accidents and fatalities on all four bar plots - a general trend can still be observed in the remaining accident data (*Figure 3*). For both pedestrian and bicycle accidents, lower percentages of total accidents were found during this time period in historical HOLC grade "A" (0.88% and 1.44%) and "B" (5.51% and 5.10%) neighborhoods compared to grade "C" (28.16% and 25.65%) and "D" (13.66% and 13.84%) neighborhoods. Similarly, for both pedestrian and bicycle fatalities, lower percentages of total fatalities were also noted in this time period for historical HOLC grade "A"

(0.42% and 0%) and "B" (4.39% and 1.16%) neighborhoods compared to grade "C" (23.43% and 24.42%) and "D" (12.51% and 12.79%) neighborhoods. These results suggest that there may be a correlation between historical HOLC redlining in 1935-1940 and recent 2020-2025 higher numbers of pedestrian and bicycle accidents and fatalities in grade "C" and "D" areas.

Data

The EJScreen data used for this assignment is from the EJScreen: Environmental Justice Screening and Mapping Tool, which no longer exists online. However, our data can be accessed through the following link: https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcIQ/view

The HOLC data used for this assignment can be found at the following Mapping Inequality website: Nelson, R. K., Winling, L, et al. (2023). Mapping Inequality: Redlining in New Deal America. Digital Scholarship Lab. Retrieved October 21, 2026, from https://dsl.richmond.edu/panorama/redlining.https://dsl.richmond.edu/panorama/redlining/data

The California Statewide Integrated Traffic Records System (SWITRS) pedestrian and bike crash data can be accessed through the Transportation Information Mapping System (TIMS) website: University of California, Berkeley Safe Transportation Research and Education Center. (n.d.). Transportation Injury Mapping System (TIMS). Retrieved October 15, 2026, from https://tims.berkeley.edu/

References

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